



EE 518: Information Theory

Course Code: EE 518

Course Name: Information Theory

L-T-P-C: 3-0-0-3

Prerequisites: IC-210 Signal and Systems, EE-304 Communication Theory or equivalent and the instructors consent

Intended for: UG/PG

Distribution: Elective

Approval: 9th Senate

Course Contents

- **Module I:** Concept of information, Information measures: Hartley measure, Shannon Entropy. [3 Lectures]
- **Basic notions:** Entropy, joint and conditional entropy, relative entropy, mutual information, KL-distance, Jensen and Log-sum inequalities. [6 Lectures]
- **Source compression:** Asymptotic Equipartition Property (AEP) and its consequences for data-compression, types of codes, Kraft inequality, optimal codes and bounds of their lengths, Huffman codes and their optimality, Shannon code, Arithmetic coding. [6 Lectures]
- **Channel coding:** the notion of channel capacity, discrete memoryless channels, channel capacity computation for elementary DMC channels, symmetric channels, jointly typical sequences, data-processing and Fano's inequalities, channel coding theorem (achievability and converse), feedback capacity, source-channel separation theorem and joint source- channel coding: multimedia communications. [9 Lectures]
- **Differential entropy:** AEP for continuous variables, joint and conditional differential entropy, relative entropy and mutual information. [3 Lectures]
- **Gaussian channel:** Coding theorem for Gaussian channels, Bandlimited channels, parallel Gaussian channels, channels with colored Gaussian noise: water-filling argument. [9 Lectures]
- **Advanced topics:** Rate-distortion theory, Network coding, Introduction to Network Information Theory. [6 Lectures]

Reference Books:

1. I. Csiszr and J. Krner, **Information Theory: Coding Theorems for Discrete Memoryless Systems**, Cambridge Univ. Press, August 2011.
2. R. G. Gallager, **Information Theory and Reliable Communication**, Wiley, 1968.
3. T. M. Cover and J. A. Thomas, **Elements of Information Theory**, 2nd Edition, Wiley, 2006.
4. D. J. C. MacKay, **Information Theory, Inference and Learning Algorithms**, Cambridge Univ. Press, 2003.